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CORE TECH

13 Technologies to watch out for

A couple of months ago we covered 5 technologies that will be making waves in 2014. Now we have a mega list with 8 additional ones to watch out for.

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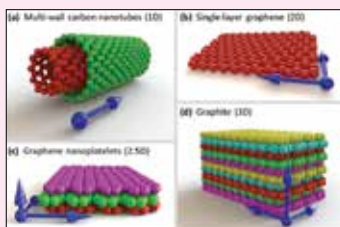
1 Brain-Computer Interface

The barrier between machine and mind is narrowing as research into brain-computer interfaces hits high gear. Researchers at Duke University have already moved to connecting the brains of two mice over the internet. The mice located in different countries were able to use this “brain net” to collaborate towards solving simple tasks for rewards. Another experiment found success when Harvard researchers connected a rat and human brain using non-invasive techniques. The most ground breaking work was done when false computer based memories were transplanted into a mouse brain at MIT. The technological implications are staggering and will completely alter the way in which we interact with computers.



2 Nanostructured Carbon Composites

Nanostructured carbon composites could hold the key to addressing three major concerns facing the world – reducing our carbon footprint, improving passenger safety in vehicles and not polluting the environment. Their use in manufacturing can reduce the weight of vehicles by more than 10 percent. This small change has serious fuel efficiency consequences and can help reduce carbon emissions. In addition to which the light weight material proves to be highly impact resistant due to its surrounding polymer matrix making vehicles safer for passengers. And most of all, it's designed to be easily reusable and not be discarded as a pollutant. A singular answer to our environmental problems with cars.



3 Adaptive Technology

Wearable technology isn't just about Google Glass and Nike+ sensors. The real emergence in wearable adaptive technology takes place in its seamless integration to our human-experience. From the use of temporary tattoos that track vitals, to foot sensors that guide the blind using vibrations and GPS – the potential for these products is more than a style statement. The most recent breakthrough in this field is the creation of the Single Pixel Contact Lens, that provides users with a Heads Up Display without any external accessory. This microscopically designed contact lens prototype is only the beginning of how technology is being adapted to the human experience to serve human needs.



4 High Density Batteries

Current lithium-ion batteries have reached their limit of energy density (energy per volume) and can't address upcoming energy concerns. New developments are underway to increase energy density - from nano-component based lithium batteries to lithium-sulphur batteries which seek to increase their limit exponentially. These new batteries will make products like emerging systems, long distance electric cars, wearable technologies and even home based solar energy generators to become the norm. This new generation of high density batteries can make even the most impoverished societies energy efficient and improve the lives of billions across the world.



5 Manufacturing at Home

Consumerism with all its benefits and flaws has remained largely unchanged due to mass production since the industrial age and. Products are created in bulk for cost effectiveness and distributed via various channels to consumers. However, the next generation of consumerism is here with additive manufacturing a.k.a. 3D printing taking hold. The ability to create complex and useful items such as decorations, crafts, tools, food and even medication via 3D printers is slowly being developed. In the emerging age of customised individualism manufacturing most of your goods from the comfort of your home will change how people consume products making the term “mass produced” obsolete.

